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The Impact of ESG Rating Disagreement on Stock Mispricing: Evidence from China

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ABSTRACT

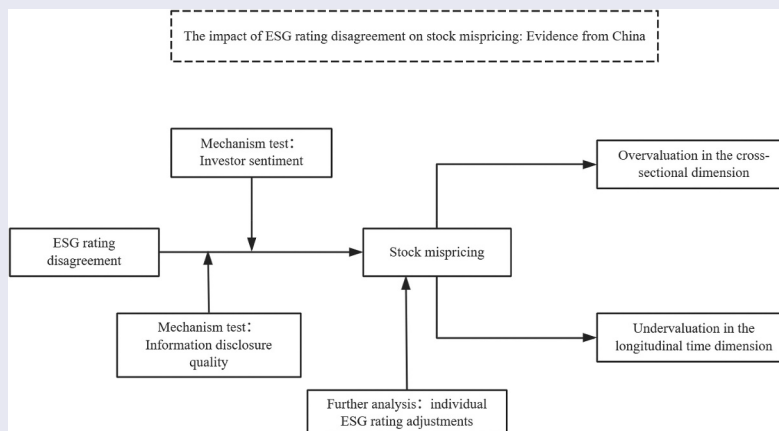
Different rating agencies often assign varying ESG ratings to the same enterprise, and the extent to which this affects the short- and long-term valuation of stock prices warrants further investigation. Using a sample of Chinese listed firms spanning 2015 to 2022, this study employs a two-way fixed effects model to investigate how ESG rating disagreement influences stock mispricing. First, this study divides stock mispricing into cross-sectional and longitudinal time dimensions. It shows that ESG rating disagreement is linked to relative overvaluation in the cross-sectional dimension and undervaluation in the longitudinal time dimension. Second, this study examines factors that might influence how ESG rating disagreement affects stock mispricing. It finds that investor sentiment and information disclosure quality are key moderators of this relationship. Third, the analysis further extends ESG rating disagreement to changes in individual agencies' ESG ratings, revealing that adjustments to a firm's specific ESG ratings also significantly affect stock mispricing. In conclusion, this study examines stock mispricing from both cross-sectional and longitudinal time dimensions, incorporating the dynamic changes in ESG ratings. These findings provide valuable insights for investors and government regulators, emphasizing the need for standardized ESG disclosure frameworks, enhanced investor education, and strengthened accountability in information reporting.

KEYWORDS

ESG rating disagreement; stock mispricing; investor sentiment; information disclosure quality

JEL

G32; G34; M21



Introduction

Sustainable development has become a priority for countries around the world. As a major carbon dioxide emitter (Kartal, Magazzino, and Pata 2024), China has formally committed to peaking its carbon emissions by 2030 and achieving carbon neutrality by 2060 (Solangi and Magazzino 2025), placing significant pressure on its renewable energy transition. To achieve these targets, China has implemented a series of legislative measures, standards, and financial investments designed to reduce pollution, preserve natural resources, and promote green innovation, thereby fostering sustainable growth (Solangi, Alyamani, and Magazzino 2024). ESG integrates environmental protection, social responsibility, and governance into business strategies, gradually becoming a critical metric for assessing corporate sustainability. This helps guide capital markets toward sustainable investment (Bolognesi and Burchi 2023). A growing number of enterprises are embedding ESG principles into multiple aspects of daily operations. Statistics from the Global Sustainable Investment Alliance (GSIA) indicate that global sustainable investment assets reached approximately USD 30.3 trillion by 2021.¹ By the end of April 2024, 1938 A-share listed companies in China had disclosed ESG-related information. At the same time, corporate ESG reporting practices have evolved from traditional social responsibility reports to ESG reports and sustainability-focused disclosures.

ESG is more than just a systematic standard for evaluating corporate integrity. It is also a strategy centered on sustainable development, which is crucial for promoting high-quality economic development. This has increasingly drawn attention from both the business and academic communities. Specifically, ESG performance mitigates managerial misconduct (F. He, Du, and Yu 2022), enhances investment efficiency (Y. Lin, Lu, and Wang 2023), and improves stock liquidity (F. He, Feng, and Hao 2023). This creates a more stable operating environment for enterprises. Furthermore, ESG performance exerts far-reaching spillover effects on green innovation (J. Li, Lian, and Xu 2023) and mitigates risk transmission within financial markets (M. Liu et al. 2023). ESG investments can provide effective protection during periods of economic uncertainty and exhibit hedging attributes (J. Wang, Wei, and Wang 2025). As an important reference for guiding ESG investment, ESG ratings have been extensively researched and published by global data companies, universities, and other institutions. However, rating results from multiple institutions for the same entity frequently lack consensus, ultimately forming ESG rating disagreement (Berg, Kölbel, and Rigobon 2022). This divergence is particularly acute within the Chinese context. Luo, Yan, and Yan (2023) point out that ESG ratings among Chinese institutions exhibit high differentiation, while M. Liu (2022) demonstrates that the correlations among six ESG rating agencies in China are mostly between 0.4 and 0.6.

To illustrate the trend of Chinese investors' attention to corporate ESG ratings and the information differences under multiple ESG rating agencies, this study compiles a total of 2.72 million investor inquiries from the Shanghai Stock Exchange e-Interaction and Shenzhen Stock Exchange Interactive Platform between 2010 and 2023. We employ an artificial intelligence large language model to identify investor inquiries and corporate responses. Figure 1(a) indicates a rapid surge in Chinese investors' attention to ESG issues after 2019, while Figure 1(b) reveals that 68.5% of investors focus on ratings from two or more rating agencies, with an average of 4.92 ESG ratings mentioned in investor inquiries. Moreover, 84.35% of investors believe the target firms' ESG ratings fall below the industry average and therefore inquire about the firms' future ESG initiatives. Furthermore, firms monitor an average of 0.25 fewer ESG ratings than investors and tend to assign systematically higher internal ESG assessments. In contrast, the ESG ratings investors cite are, on average, two levels lower than the firms' own subjective assessments.² In general, the differences in ESG ratings for listed companies primarily manifest in three aspects: First, firms believe that the evaluation indicators and weighting designs of different rating agencies are highly subjective and lack unified standards, resulting in significant discrepancies in rating outcomes among agencies. Second, agency ratings depend heavily on publicly disclosed information, which may deviate from firms' actual conditions and thereby reduce rating accuracy. Third, many firms note insufficient communication from rating agencies, leading to assessments that fail to accurately reflect their operational realities.

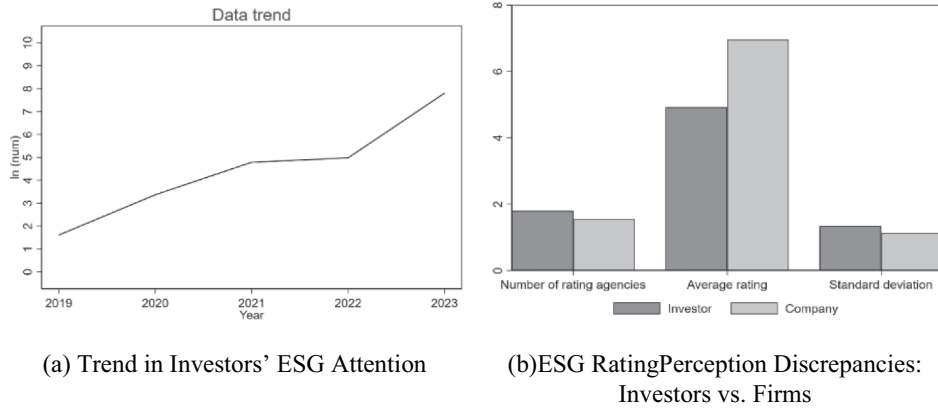


Figure 1. Analysis of ESG attention dynamics and perception Differences. *Note:* Based on investor inquiry data from 2010 to 2023. Figure 1(a) shows the continuous rise in Chinese investors' ESG attention, while Figure 1(b) exhibits disparities in the number of ESG rating agencies tracked by investors and firms, along with variations in average rating scores and standard deviations.

According to the Efficient Market Hypothesis, market information can be efficiently and accurately reflected in stock prices (Ruan, Li, and Huang 2024). Enhancing pricing efficiency is a cardinal function of capital markets. The severity of stock mispricing to some extent also reflects market pricing efficiency (Y. Wang, Tang, and Li 2025). Specifically, stock mispricing denotes the deviation between a stock's market price and the intrinsic value derived from fundamentals (Gao, Cui, and Xu 2025). Severe distortions can compromise capital allocative efficiency and potentially precipitate systemic financial instability (Wu et al. 2024). Emerging capital markets often exhibit the phenomenon of relatively widespread stock mispricing (W. Li, Yang, and Yin 2022), with stock mispricing in the Chinese market being particularly pronounced (X. Lin, Zhu, and Meng 2023). As divergence among ESG ratings intensifies, its impact on ESG investment has drawn widespread attention. Emerging literature has explored the relationship between ESG rating disagreement and asset pricing, but there is still no clear agreement. Some studies indicate that ESG rating disagreement can provide the market with more comprehensive insights into the sustainable development performance of enterprises. By offering analysts different assessment perspectives and viewpoints, it enriches the data sources for analysts to analyze and evaluate corporate earnings. The heterogeneity of ESG ratings can reveal a company's ESG information from multiple perspectives and provide a comprehensive understanding of its operational status. However, other research suggests that ESG rating disagreement causes information redundancy, increasing the difficulty of interpreting ESG information of enterprises. The presence of ambiguity and excessive information can lead to information overload. ESG rating disagreement causes market participants to have different views on a firm's ESG performance, impeding the ability of market participants to obtain consistent information to measure its true potential.

Given this context, this paper examines the impact of ESG rating disagreement on stock mispricing among Chinese listed companies from 2015 to 2022. In this regard, this study selects China as the research background primarily for the following reasons. As the world's second-largest economy and the largest emerging market, China plays a pivotal role in worldwide environmental governance (Zhu et al. 2025). The rapid development of its own ESG practices provides plenty of company data for research. Concurrently, China's ESG rating system has its uniqueness. The development of the relevant ESG rating system in China remains relatively nascent (Q. Chen and Li 2024), and the lack of standardization leads to significant differences in corporate ratings (Zhu et al. 2025). Meanwhile, unlike the "top-down" frameworks in developed countries, China's ESG development follows a "bottom-up" approach. It primarily exerts regulatory pressure through soft regulatory mechanisms to drive corporate sustainability (Y. Li et al. 2024). Furthermore, the developmental quality of China's

capital market has drawn attention from global researchers and investors (Ma, Guo, and Li 2024). Compared with developed countries, China's capital market has deficiencies in terms of functionality and regulatory framework (W. Li and Zheng 2024). Moreover, retail investors dominate the Chinese stock market. These individual investors struggle to engage effectively with companies through voting or proposals (Ruan and Yang 2025). The prevalence of retail traders exacerbates market irrationality, further contributing to stock mispricing (W. Li and Zheng 2024). In China's emerging market, addressing the informational needs of retail investors has become a pressing issue (Ruan and Yang 2025), especially facing more severe challenges in the context of stock mispricing. Consequently, the unique circumstances of China are well-suited for studying how ESG rating disagreement affects stock mispricing.

This paper primarily investigates how ESG rating disagreement influences stock mispricing and the mechanisms behind this effect. Basic regression results show that increased ESG rating disagreement contributes to overvaluation in the cross-sectional dimension and undervaluation in the longitudinal time dimension. Mechanism analysis shows that investor sentiment and information disclosure quality significantly moderate this relationship. In addition, individual ESG rating adjustments can also impact stock mispricing. The marginal contributions are primarily reflected in the following aspects. First, with the growing global focus on sustainable development goals, a thorough analysis of ESG rating disagreement helps achieve social and environmental benefits. It also promotes economic development toward a green and sustainable direction. Investigating whether ESG rating disagreement at the corporate level can influence stock mispricing holds significant practical implications for achieving the “Dual Carbon” targets and promoting sustainable economic development. Second, we further enrich the micro-evidence linking ESG rating disagreement to market performance. Research findings indicate that higher ESG rating disagreement causes overvaluation at the cross-sectional dimension and undervaluation at the longitudinal time dimension, filling a gap in existing research on this relationship. Our results support the “noise effect” hypothesis and offer insights into how markets process the complexity of ESG signals. Third, this study introduces moderating variables to refine the impact mechanisms. Building upon the analysis of the direct effect of ESG rating disagreement on stock mispricing, we also explore the moderating effects of investor sentiment and information disclosure quality. This paper supplements the mechanism literature on ESG rating disagreement from the perspective of external environmental regulation, which is helpful for understanding the economic consequences of ESG rating disagreement. Furthermore, this study investigates how the direction of individual ESG rating adjustments relates to stock mispricing. Fourth, China is a major emerging market nation with immense development potential and opportunities. The Chinese market mirrors conditions in other emerging economies. Through research on China's ESG rating phenomenon, this study provides valuable insights for other emerging markets and offers beneficial suggestions for regulators, policymakers, and market participants in China and other countries and regions.

The structure is organized as follows: Section “Literature Review” reviews the relevant literature; Section “Theoretical Background and Hypothesis Development” presents the theoretical analysis and hypotheses; Section “Research Design” discusses the research design and variable construction; and Section “Empirical Analysis” reports the empirical results. Finally, Section “Discussion” discusses the findings, and Section “Conclusions and Policy Implications” presents the conclusions and policy implications.

Literature Review

With the acceleration of industrialization and urbanization, issues including resource depletion and ecological destruction have gradually emerged. In response, the ESG framework has emerged, advocating that companies should consider environmental, social, and governance performance alongside financial objectives. By integrating ESG factors into investment choices, firms can not only follow the mandatory policy standards of the government, but also meet public expectations for sustainable development (Z. Wang and Zheng 2025). Existing research focuses on the influencing factors of ESG

and its economic consequences, confirming that ESG practices positively drive corporate high-quality development and enhance capital market performance. In addition to internal factors such as characteristics of executive directors and financial flexibility (H. Li et al. 2025; Lv et al. 2025; Y. Zhang and Li 2025), external factors including industry competition, digitalization levels, media coverage, and green finance also significantly influence corporate ESG ratings (P. Li, Li, and Wu 2025; Seng and Zhang 2025; Q. Wang and Xu 2025; Z. Wang and Zheng 2025). At the company level, good ESG performance can effectively enhance corporate value and financial performance (Aydoğmuş, Gülay, and Ergun 2022), reduce systemic tail risk exposure, and promote green innovation capabilities (X. Liu, Wu, and Li 2025; J. Wang et al. 2023), ultimately driving sustainable corporate development. At the level of capital market performance, Chowdhury et al. (2021) show that higher levels of CSR information alleviate information disadvantages for foreign companies in the U.S. capital markets, ultimately reducing idiosyncratic stock volatility and enhancing liquidity. Furthermore, ESG assets help protect portfolio performance against extreme market volatility and tail risks (Ben Ameer, Ftiti, and Louhichi 2025), and affect stock liquidity and returns (Fan et al. 2024; K. Wang et al. 2023).

However, as ESG investing grows, significant discrepancies exist between ESG ratings from different agencies, and overall confusion is widespread. This ambiguity primarily comes from the inherent complexity of compiling ESG assessments and validating data sources (Rossi, Byrne, and Christiaen 2024). Any attempt to quantify ESG performance needs to deal with the challenges of unstructured methodologies and incomplete, opaque ESG data (Avramov et al. 2022). Different agencies employ varying data sources, methodologies, and rating models when assessing ESG performance (Guo et al. 2024), which leads to different ratings for the same firm. Furthermore, companies still lack a unified framework for reporting ESG information. Some sub-indicators exhibit a bias toward “reporting good news but not bad news” and are subject to “selective disclosure.” In China, these issues with ESG ratings are even more noticeable. Firstly, as the development of China’s ESG rating system progresses, rating agencies exercise greater discretion in interpreting corporate ESG performance. Secondly, both domestic and foreign rating agencies coexist in the Chinese market. Their growing numbers further intensify ESG rating disagreement (X. Liu, Yang, et al. 2024). This divergence complicates investment decisions and poses greater challenges for communication between corporate management and stakeholders. The controversy surrounding corporate ESG ratings has intensified, significantly impacting stock price performance and capital market efficiency. It has garnered significant attention from investors and become the focus of academic research. Stock mispricing is the deviation between a stock’s price and its intrinsic value (Sakaki, Jory, and Jackson 2021). This deviation affects capital market efficiency and hinders the development of markets (W. Li and Zheng 2024). Azevedo and Müller (2024) provide compelling evidence demonstrating that global stock markets fail to achieve efficient pricing. Furthermore, D. Liu, Gu, and Lung (2016) examine pricing errors in China’s stock market, indicating that the overvaluation of the market stems from two main factors: the dominance of individual investors and the high percentage of state-owned or government-related enterprises. Existing literature attributes stock mispricing to investor behavior (Miwa 2016; B. Yang, Ye, and Ma 2022) and institutional factors such as capital market registration system reforms (Y. Wang, Tang, and Li 2025). Additionally, clean energy indices, ESG and greenwashing factors (Fang, Fu, and Hashemishahraki 2025; Z. Yang et al. 2024; X. Lin, Zhu, and Meng 2023) are also regarded as important determinants of stock mispricing.

It is crucial to clarify the economic impact of ESG rating disagreement on capital markets. Recently, an emerging strand of literature has explored the influence of ESG rating disagreement on corresponding asset pricing, but no consistent conclusions have yet been reached. Some studies argue that ESG divergence causes information overload, exacerbating informational asymmetry between investors and firms. This prevents investors from accessing effective ESG information for their own decisions, complicates the interpretation of corporate ESG information, and exerts a “noise effect.” Specifically, the widespread divergence in ESG scores is not conducive to the accurate valuation of ESG risks and weakens the connection between investors’ ESG preferences and well-performing ESG stocks

(Anselmi and Petrella 2025). ESG rating disagreement increases the uncertainty in ESG information, reducing the information content of stock prices and stock liquidity (K. Hu, Li, and Li 2024; X. Zhang, Sun, and Gao 2025). This disagreement reduces stock excess returns (H. Wang et al. 2024) and capital market efficiency (X. Lin, Zhu, and Meng 2023; Ruan, Li, and Huang 2025). Additionally, a higher ESG score divergence may amplify investors' differing perceptions of corporate ESG performance, ultimately increasing idiosyncratic return volatility (X. Liu, Yang, et al. 2024). Another body of research suggests that ESG rating disagreement can provide the market with more comprehensive insights into corporate sustainable development performance. By offering analysts different assessment perspectives and incorporating additional information into ratings, this divergence helps investors understand enterprises from multiple dimensions, thereby generating an "information effect." Specifically, the heterogeneous results of ESG ratings can reveal ESG information from multiple dimensions. As investors' learning effects strengthen, this information is more effectively incorporated into stock prices. Ultimately, such inconsistencies help improve corporate pricing efficiency (Yin et al. 2025). Furthermore, ESG rating disagreement positively incentivizes enterprises to voluntarily disclose information, lowering the risk of future stock price crashes (Ju, Fang, and Shen 2025).

A review of the existing literature reveals that, first of all, the current research on ESG focuses on influencing factors and their economic consequences, affirming from all aspects the positive role of ESG practices in promoting corporate development and capital market performance. With the development of ESG investment, the differences in ESG ratings have become increasingly pronounced. ESG ratings from different agencies exhibit significant discrepancies, attributed to varying methodologies and the lack of standardized ESG criteria. This divergence complicates investment decisions and also exacerbates communication challenges between company managers and stakeholders. Secondly, it is also important to clarify the economic effects of ESG rating disagreement. As ESG divergence increases, controversies surrounding corporate ESG ratings are becoming increasingly intense, attracting investor attention and exerting significant economic impacts on company stock performance. Third, emerging literature explores how ESG rating disagreement influences corresponding asset pricing but has yet to reach consistent conclusions. Some studies indicate that ESG rating disagreement causes information overload, failing to provide effective ESG information for investor decision-making. Other studies argue that rating disagreement can provide the market with more comprehensive insights into corporate sustainable development performance. This divergence incorporates more information into ESG ratings and helps investors understand corporate performance from multiple aspects. Furthermore, although some literature has explored the impact of ESG rating disagreement on stock efficiency, most studies indicate that it reduces stock efficiency. However, these studies have not distinguished specific characteristics associated with stock market pricing efficiency from cross-sectional and longitudinal time dimensions. Specifically, in the short term, ESG rating disagreement may be more likely to cause market fluctuations, while in the long term, as information is gradually digested and market mechanisms self-adjust, the recovery of stock pricing efficiency may be relatively slow. Our study explores the influence of ESG rating disagreement on stock mispricing. Meanwhile, this research attempts to introduce investor sentiment and information disclosure quality factors to elucidate the mechanisms linking ESG rating disagreement to stock mispricing.

Theoretical Background and Hypothesis Development

First, the Efficient Market Hypothesis posits that in an ideal market, stock prices should accurately reflect the value of the underlying assets, thereby preventing investors from achieving excess returns (K. Li, Xu, and Zhu 2014). Second, Information Asymmetry Theory indicates that non-financial disclosures can reflect a company's commitment to environmental and social goals, thereby helping to reduce information asymmetry. However, ESG rating disagreement creates more noise, thereby impeding the ability of external investors to assess the fundamental value of target firms (X. Lin, Zhu, and Meng 2023). Furthermore, the concept of limited rationality indicates that participants in the capital market are not fully rational (Wei and Sun 2024). These investors are more easily misled by

fabricated or exaggerated environmental, social, and governance information disclosures. Due to psychological cognitive biases, investors often pay excessive attention to immediate information when faced with different types of information, which can easily lead to the phenomenon of “excessive rises and falls” in stock prices.

Disagreement among ESG rating agencies can lead to more noise in the information environment. On the one hand, conflicting rating signals from ESG rating agencies make it harder for investors to figure out a firm’s true ESG performance. This amplifies information noise, ultimately increasing investors’ information processing costs. Investors influenced by the representativeness heuristic tend to overlook the fact that historically high profit growth is unlikely to be sustainable, ultimately overestimating a company’s value and becoming disappointed when anticipated future profit growth fails to materialize. Therefore, following a series of positive announcements, investors become overly optimistic, leading to an overreaction that ultimately pushes stock prices to excessively high levels (Barberis, Shleifer, and Vishny 1998). Thus, the noise of ratings can cause investors to overreact in the short term and make erroneous investment decisions. This leads to a deviation in stock prices. On the other hand, ESG rating disagreement implies that rating agencies lack consensus on the ESG-related risks, opportunities, and long-term value of companies, thereby increasing information uncertainty for investors. ESG rating disagreement increases market uncertainty. If biased rating information spreads to the market, it will lead investors to make irrational decisions. Notably, this phenomenon is particularly pronounced in China due to the severe short-selling restrictions inherent (B. Yang and Ma 2021) and the speculative demands of individual investors (Shi et al. 2023). Therefore, the pricing in the stock market is more susceptible to ESG rating disagreement, leading to a greater deviation between stock prices and fundamental value.

Furthermore, in markets with short-sale restrictions, stock prices often only reflect the expectations of optimistic investors. Consequently, stocks of highly watched enterprises face upward price pressure; specifically, high-rated enterprises receiving more investor attention experience upward price pressure driven by short-term net buying behavior. This behavior tends to create a positive feedback mechanism between rising stock prices and investor attention, leading to significant deviations in stock prices from the firm’s intrinsic value. On the one hand, in markets with high short selling costs, asset prices are primarily dominated by optimistic investors, who often reflect more optimistic valuation prices. Miller (1977) found that when investors have divergent opinions about an asset’s future value, asset prices are established by the expectations of optimistic investors. Pessimistic investors cannot fully express their views through short-selling and reflect them in prices, leading to asset overvaluation. At this time, the presence of many well-informed investors will prevent securities from being severely undervalued, but some securities may also have their prices pushed to excessively high levels by a well-informed minority. Although different opinions exist in the market, it still tends to suppress pessimistic selling, generating optimistic prospects and further increasing prices. On the other hand, when rating agencies disagree, current ratings fail to accurately reflect enterprises’ ESG performance and provide effective a sound basis for investor decision-making. Firms subject to elevated ESG rating divergence exhibit greater performance uncertainty and information asymmetry, often enhancing investor optimism and causing investors to be overly optimistic about enterprises’ future performance while blindly following trends. However, in the long term, as companies release more ESG reports, receive more ESG awards and recognition, and media continue to report on companies’ ESG social news, market sentiment will tend toward rationality. The heterogeneous beliefs regarding ESG rating disagreement faced by investors will continuously decrease. At this point, the overvaluation previously caused by noise effects and excessive optimism loses support, prompting stock prices to correct downward toward their fundamental value. If the correction is severe, prices may even fall below their fundamental value, resulting in a longitudinal undervaluation.

In summary, when investors encounter ESG rating disagreement, they tend to select the highest ESG rating to represent the company’s true ESG performance. This will lead investors to overestimate the firm’s ESG performance and underestimate future ESG risks, thereby causing stock overvaluation. However, in the long term, due to delayed information verification, this overvaluation becomes

unsustainable. Over time, the market gradually identifies the “noise” in the divergence, and risk repricing leads to valuation corrections, thereby reducing long-term stock returns. These irrational investors are often overconfident and usually trust their own judgments more than they rationally utilize the market information. This will lead to overvaluation of the current stock situation, and further causes a decline in future earnings (J. Wang et al. 2024). Additionally, under conditions of information asymmetry and risk, investors underprice high-quality enterprises in order to avoid potential losses (Healy and Palepu 2001). Theoretically, the higher the ESG rating disagreement, the more likely long-term stock prices are to be undervalued. An excessive focus on short-term ESG performance may overlook long-term financial fundamentals or market dynamics, contributing to the undervaluation of long-term stock prices.

Building upon the previous analysis, this study posits Hypothesis H1: As ESG rating disagreement increases, it engenders cross-sectional overvaluation of stock prices while simultaneously precipitating longitudinal time undervaluation. Specifically, the higher the ESG rating disagreement, the higher the company’s stock valuation compared to contemporaneous companies in the same industry, and the lower the valuation compared to long-term prices.

Research Design

Sample and Data

To explore how ESG rating disagreement affects stock mispricing, this study selects listed companies in China from 2015 to 2022 as the research sample. The ESG rating data are sourced from the Bloomberg and WIND databases, while stock trading, market trading, and firm-level data are obtained from the CSMAR and WIND databases. The sample is filtered through the following steps: (1) Exclude the enterprise samples in the financial industry and the real estate industry; (2) Exclude enterprise samples classified as ST or *ST; (3) Exclude firms in the year of their initial public offering (IPO). Additionally, continuous variables in the research sample are winsorized at the 1% and 99% levels.

Variables

Dependent Variable: Stock Mispricing

Rhodes-Kropf, Robinson, and Viswanathan (2005) proposed to measure stock mispricing by decomposing the market-to-book ratio into mispricing and intrinsic value. This model is based on a retrospective benchmark approach using accounting data (Shan and Chen 2025), which helps avoid the risk of capturing company growth opportunities instead of management timing choices (Hussain et al. 2023). Furthermore, the model imposes fewer constraints and allows for greater heterogeneity in discount rates across industries (W. Xue, He, and Yun 2025). A substantial body of literature has measured stock mispricing indicators based on this method (Aabo, Pantzalis, and Park 2017; Andreou, Lambertides, and Panayides 2021; Bonaimé, Öztekin, and Warr 2014; Pantzalis and Park 2009). Therefore, this study employs Equation (1) to examine the level of stock mispricing:

$$\text{Market to Book} \equiv \text{Market to Value} \times \text{Value to Book} \quad (1)$$

Specifically, *Market*, *Book*, and *Value* represent the market value, book value, and intrinsic value of equity, respectively. If *Value* has an accurate value measure, the right side of Equation (1) is partitioned into two components. Among these, *MarkettoValue* measures the deviation of price from true value, while *ValuetoBook* represents the true book value, measuring growth opportunities in a way unaffected by valuation errors. Taking the logarithm of Equation (1) leads to Equation (2):

$$m - b = (m - v) + (v - b) \quad (2)$$

Furthermore, by introducing a linear function that estimates coefficient *a* based on financial information, Equation (2) can be reformulated as Equation (3):

$$m_{it} - b_{it} = m_{it} - v(\theta_{it}; a_{jt}) + v(\theta_{it}; a_{jt}) - v(\theta_{it}; a_j) + v(\theta_{it}; a_j) - b_{it} \quad (3)$$

Specifically, m_{it} and b_{it} denote the natural logarithm of total market value and book value. $v(\theta_{it}; a_{jt})$ indicates the fundamental value for firm i in year t and industry j , that is, the company value determined by company-specific factor set θ_{it} for company i in period t without considering industry valuation bias. $v(\theta_{it}; a_j)$ represents the industry fundamental value of enterprise i given industry j , excluding the time-series characteristics of industry valuation. Among these, $m_{it} - v(\theta_{it}; a_{jt})$ measures the deviation between a company's stock market value and the estimated value of contemporaneous companies in the same industry, representing cross-sectional dimension mispricing. $v(\theta_{it}; a_{jt}) - v(\theta_{it}; a_j)$ measures the discrepancy between a firm's current industry estimated value and long-term industry estimated value by measuring the differences arising when contemporaneous multiples differ from long-term multiples, representing longitudinal time dimension mispricing.³

Explanatory Variable: ESG Rating Disagreement

This study adopts the methodology proposed by C. Zhang et al. (2024), utilizing the four most prominent ESG rating systems to measure ESG rating disagreement. Specifically, it adopts four types of rating indicators: Huazheng, Wind, SynTao Green Finance, and SusallWave ESG ratings; it then assigns values to the rating results and calculates their standard deviation as the measurement indicator of ESG rating disagreement. Furthermore, following the research of Ling, Xia, and Liu (2024), this study selects six types of indicators: Huazheng, Wind, SynTao Green Finance, SusallWave, Bloomberg, and FTSE Russell ESG ratings; it then assigns values to the rating results and calculates the standard deviation for robustness tests. Additionally, this study employs multidimensional approaches to compare ESG ratings, including Principal Component Analysis (PCA), range, interquartile range (IQR), and coefficient of variation (CV), to further measure ESG rating disagreement.

Moderating Variables

The first moderating variable is Investor Sentiment. Following Rhodes-Kropf, Robinson, and Viswanathan (2005), this study decomposes Tobin's Q into intrinsic value and market mispricing. Considering industry heterogeneity and market fluctuations, a cross-sectional regression is conducted for all firms within each industry annually, as shown in Equation (4):

$$Q_{it} = \beta_0 + \beta_1 size_{it} + \beta_2 lev_{it} + \beta_3 roa_{it} + \varepsilon_{it} \quad (4)$$

Rhodes-Kropf, Robinson, and Viswanathan (2005) identify firm size, leverage, and profitability as the pivotal determinants of intrinsic value. In this study, the fitted values from the above model are used as a proxy for investment opportunities in the empirical research. The residuals are then Z-standardized to derive a proxy variable for the sentiment indicator, which represents individual stock-level investor sentiment, referred to as *Sen*.

The second moderating variable is Information Disclosure Quality. Since the KV index integrates mandatory and voluntary reporting to proxy for the substantive effectiveness of corporate disclosure, this study follows the method in Kim and Verrecchia (2001) to measure information disclosure quality using the KV value. When listed companies commit to timely disclosure of information within a specific performance range, they delay disclosure if performance falls below that range. During the delayed disclosure, market makers can infer investors' private information about company value through trading volume, while at this time stock returns depend on trading volume. In this context, the trading volume slope coefficient decreases as the company's committed timely disclosure range expands. Therefore, a smaller KV value indicates that the listed company commits to a larger performance range for timely disclosure with more adequate information disclosure. Accordingly, lower KV values represent higher information disclosure quality. The model is specified in Equation (5):

$$\ln|\Delta P_t/P_{t-1}| = \alpha + \beta \times (vol_t - vol_0) + \mu_i \tag{5}$$

In this context, P_{t-1} denotes the closing stock price on day $t - 1$; ΔP_t signifies the difference between P_t and P_{t-1} ; vol_t and vol_0 represent the trading volume on day t and the average daily trading volume for the year, respectively. Through regression of Equation (5), β is obtained, and finally the KV value is calculated.

Control Variables

Consistent with prior literature, this study incorporates control variables related to a firm’s financial condition, governance structure, and ownership nature. Specifically, the control variables are firm age, total asset turnover, profitability, growth ability, fixed asset status, independent director proportion, ownership concentration, annual turnover rate and institutional ownership ratio. Table 1 presents detailed definitions of the primary variables.

Model Specification

Based on the method in J. Zhang et al. (2025) and H. Hu, Jia, and Yang (2025), this study utilizes a two-way fixed effects model to investigate the influence of ESG rating disagreement on stock mispricing. First, this model mitigates omitted variable bias stemming from time-invariant firm attributes or common temporal shocks. Second, this model is suitable for panel data analysis. By utilizing cross-sectional and time dimensions, it provides more robust and reliable estimates (Duan et al. 2025). The model, as shown in Equations (6) and (7), accounts for individual company characteristics that remain

Table 1. Definition of primary variables.

Variable Type	Name of Variables	Variable Symbol	Variable Description
Dependent Variable	Stock mispricing	$MISP_{cross}$	Firm-level mispricing
		$MISP_{time}$	Industry-level mispricing
Independent Variable	ESG rating disagreement	ESG_dis	Standard deviation of ESG rating scores from Huazheng, Wind, SynTao Green Finance, and SusallWave
		ESG_dis6	Standard deviation of ESG rating scores from Huazheng, Wind, SynTao Green Finance, SusallWave, Bloomberg and FTSE Russell
		$ESGD$	ESG rating disagreement among four types of indicators under Principal Component Analysis
		$ESGrange4$	ESG rating disagreement among four types of indicators under range method
		$ESGdis$	ESG rating disagreement among four types of indicators under interquartile range
		$Disagreement$	ESG rating disagreement among four types of indicators under coefficient of variation method
Moderating Variable	Investor sentiment	Sen	Individual stock investor sentiment
	Information disclosure quality	KV	KV index
Control Variable	Firm age	Age	The logarithm of the firm age
	Total asset turnover	ATO	The ratio of operating revenue to total assets
	Profitability	ROA	Return on assets
	Growth ability	$Growth$	Total asset growth rate
	Fixed asset status	FAS	Fixed asset ratio
	Independent director proportion	$Indep$	The ratio of independent directors to total board members
	Ownership concentration	$Top3$	The percentage of ownership of the top three shareholders
	Annual turnover rate	$Turnover$	Annual sum of daily stock turnover rates (tradable shares)
	Institutional ownership ratio	$Investor$	The proportion of shares owned by institutional investor

constant over time and general time-specific shocks that affect ESG rating disagreement. Finally, this study employs firm-level clustered standard errors in regression analysis to mitigate firm-level heterogeneity and improve the accuracy of results.⁴

$$MISP_{crossi,t} = \beta_0 + \beta_1 ESG_dis_{i,t} + \beta \text{ Control}_{i,t} + v_i + \gamma_t + \varepsilon_{i,t} \quad (6)$$

$$MISP_{timei,t} = \beta_0 + \beta_1 ESG_dis_{i,t} + \beta \text{ Control}_{i,t} + v_i + \gamma_t + \varepsilon_{i,t} \quad (7)$$

In Equations (6) and (7), $MISP_{crossi,t}$ represents cross-sectional dimension mispricing, $MISP_{timei,t}$ represents longitudinal time dimension mispricing, $ESG_dis_{i,t}$ represents ESG rating disagreement, β_1 serves as the core parameter quantifying how ESG rating disagreement influences stock mispricing. Additionally, $\text{Control}_{i,t}$ represents a set of control variables, v_i and γ_t denote firm and time fixed effects, respectively, and $\varepsilon_{i,t}$ represents the random error term.

Furthermore, we address potential endogeneity concerns. In the next section, this study also employs independent variables that are lagged one period, the instrumental variable method, a test for omitted variable bias, and the treatment effects model to mitigate endogeneity problems. Among these, the approach using independent variables that are lagged one period re-estimates the model by replacing independent variables with their first-order lags, which helps mitigate endogeneity due to reverse causality. The instrumental variable method effectively addresses endogeneity caused by reverse causality and omitted variables by selecting exogenous variables that are highly correlated with ESG rating disagreement but unrelated to the random error term. The test for omitted variable bias conducts additional robustness tests for omitted variable bias by performing coefficient stability analysis using boundary testing methods. The treatment effects model estimates a selection equation to compute the Inverse Mills Ratio (IMR), incorporating it into the main model to rectify sample selection bias.

Empirical Analysis

Descriptive Statistics

Table 2 reports the summary statistics for the primary variables. Among them, the stock mispricing indicator $MISP_{cross}$ ranges from -0.910 to 1.282 , indicating large deviations in stock mispricing and frequent mispricing phenomena in the market. $MISP_{time}$ ranges from -0.427 to

Table 2. Summary statistics for the primary variables.

	N	Mean	SD	Min	Max
$MISP_{cross}$	20626	-0.011	0.445	-0.910	1.282
$MISP_{time}$	20626	0.160	0.306	-0.427	0.966
ESG_dis	20626	0.800	0.741	0.000	2.828
ESG_dis6	20626	0.931	0.717	0.000	2.828
$ESGD$	20626	0.113	0.086	0.000	0.311
$ESGrange4$	20626	1.347	1.261	0.000	4.000
$ESGdis$	20626	0.505	0.586	0.000	2.121
$Disagreement$	20626	0.075	0.084	0.000	0.424
Sen	20303	0.064	1.115	-1.455	5.077
KV	19990	0.408	0.224	0.026	1.066
Age	20626	2.932	0.302	2.079	3.526
ATO	20626	0.620	0.397	0.094	2.442
ROA	20626	0.052	0.041	0.002	0.211
$Growth$	20626	0.179	0.302	-0.180	1.966
FAS	20626	0.206	0.151	0.003	0.672
$Indep$	20626	0.378	0.054	0.333	0.571
$Top3$	20626	0.484	0.152	0.173	0.856
$Turnover$	20626	6.282	5.143	0.554	27.488
$Investor$	20626	0.430	0.250	0.003	0.913

0.966, indicating heterogeneity in industry-level stock mispricing. Regarding ESG rating disagreement, the mean of ESG_dis is 0.800. When considering disagreement indicators from more ESG rating agencies, the mean of ESG_dis6 rises by 16.38%. The diversity of ESG rating agencies actually creates more noise for the market, aligning with the statistical evidence detailed in the stylized facts section of this study.

Tests of the Dependent Variable

Adopting the method of Rhodes-Kropf, Robinson, and Viswanathan (2005), this study regresses market value on book value, net profit, and leverage at each industry-year level; consequently, it is necessary to verify the distribution and stability of model residuals. If residuals are highly correlated with known market factors, they may not represent pure pricing errors. Therefore, this study uses residual diagnostics to supplement the analysis, and it reports correlation between valuation residuals and key market factors to confirm that residuals truly reflect pricing deviations. Specifically, Table A1 in Appendix A provides the residual analysis of mispricing variables, while Table A2 shows the correlation analysis between these residuals and key market factors.

In the residual analysis, the unit root test of panel data is applied to ensure the stationarity of the data. Compared to other commonly used panel unit root test methods, the Fisher test allows for different autoregressive coefficients across different cross-sections and permits unbalanced panels. The results show that the accompanying probabilities of the inverse chi-square, inverse normal, inverse logistic, and modified inverse chi-square statistics are consistently 0.000, indicating that the residuals are stationary. Additionally, the p-values derived from the White test exceed the 0.10 threshold, thereby precluding the presence of significant heteroscedasticity and validating the distributional stability of the model residuals.

Moreover, this study conducts a correlation analysis between residuals (*Residuals*) and known market factors. Specifically, excess stock returns and stock price volatility are selected as representative indicators of market factors. First, following the research of Ye and Xu (2013), the market adjustment method is used to calculate excess stock returns (V), with the measurement window extending from May 1 of the current year through April 30 of the subsequent year.⁵ This specific calculation is shown in Equation (8):

$$V_{it} = \prod_{j=(t,5)}^{j=(t+1,4)} (R_{ij} + 1) - \prod_{j=(t,5)}^{j=(t+1,4)} (Rm_{ij} + 1) \quad (8)$$

Specifically, R_{ij} denotes the monthly stock return of company i in month j , adjusted for cash dividend reinvestment, and Rm_{ij} represents the market return rate of the specific market segment of the stock exchange to which company i belongs in month j .

Second, as an important determinant of macroeconomic stability and enterprise development (Zheng et al. 2025), stock price volatility reflects market uncertainty regarding prospective price trajectories and investors' risk expectations (Lei and Zheng 2025). Based on this, following the work of Su (2015), L. Liu, Wu, and Peng (2025), this paper employs three methods to simultaneously measure stock price volatility: the logarithm of the annualized daily, weekly, and monthly return standard deviations ($CRT1$, $CRT2$, $CRT3$). The formula is shown in Equation (9):

$$CRT_{ijt} = \ln \left[\sqrt{\frac{1}{T} \sum_{t=1}^T \left(r_{ijt} - \frac{1}{T} \sum_{t=1}^T r_{ijt} \right)^2} \right] \quad (9)$$

Where r_{ijt} denotes the return of company i on day (week, month) t within year j , and T represents the total number of days (weeks, months) within each accounting year. Empirical analysis shows that the correlation coefficients between *Residuals* and V , $CRT1$, $CRT2$, and $CRT3$ are all below 0.3, suggesting that there is no high correlation with known market factors.

Baseline Regression Results

Table 3 reveals the impact of ESG rating disagreement on stock mispricing. Columns (1) and (2) report a positive coefficient for *ESG_dis*, suggesting that ESG rating disagreement can effectively cause stock price overvaluation at the cross-sectional dimension level. That is, as ESG rating disagreement increases, the firm's stock valuation becomes higher when compared horizontally with contemporaneous companies in the same industry. This study further examines the valuation level in the time dimension. Columns (3) to (4) display a significantly negative coefficient for *ESG_dis*, implying that ESG rating disagreement results in stock price undervaluation at a longitudinal time dimension level. That is, as ESG rating disagreement increases, the firm's stock valuation becomes lower when compared longitudinally with long-term valuation. This is because, when confronted with ESG rating disagreement, investors exhibiting optimistic bias anchor to the highest score as the proxy for the company's intrinsic ESG performance, leading to an overestimation of the current stock valuation. However, as time progresses, the overvaluation previously based on divergence noise effects and excessive optimism loses support. Consequently, stock prices correct downward and return to fundamental value, subsequently leading to underestimation of future stock prices. The empirical results support Hypothesis 1.

Robustness Tests

Alternative Measure of the Independent Variable

In the robustness tests, this study first employs *ESG_dis6* to substitute the original explanatory variable, re-estimates the model, and obtains empirical results as shown in Table 4. Among these, the coefficient of *ESG_dis6* in Column (1) is positive, while the coefficient in Column (2) is negative. The results suggest that higher ESG rating disagreement is linked to higher stock valuation compared

Table 3. Baseline model results of ESG rating disagreement and stock mispricing.

	(1)	(2)	(3)	(4)
	<i>MISP_{cross}</i>	<i>MISP_{cross}</i>	<i>MISP_{time}</i>	<i>MISP_{time}</i>
<i>ESG_dis</i>	0.021*** (4.22)	0.024*** (5.11)	-0.029*** (-8.61)	-0.028*** (-8.23)
<i>Age</i>		0.204** (2.22)		-0.059 (-0.97)
<i>ATO</i>		0.048** (2.34)		-0.022** (-2.06)
<i>ROA</i>		-1.177*** (-9.83)		0.605*** (9.46)
<i>Growth</i>		0.043*** (4.98)		-0.008* (-1.72)
<i>FAS</i>		0.182*** (3.20)		0.043 (1.35)
<i>Indep</i>		0.190** (2.28)		0.037 (0.73)
<i>Top3</i>		-0.604*** (-7.30)		-0.043 (-1.02)
<i>Turnover</i>		0.012*** (18.09)		0.006*** (18.02)
<i>Investor</i>		1.144*** (20.17)		0.170*** (6.39)
<i>Cons</i>	-0.028*** (-6.88)	-0.991*** (-3.51)	0.183*** (67.38)	0.226 (1.23)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	20626	20626	20626	20626
Adj R ²	0.6297	0.6684	0.7480	0.7585

Note: The t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 4. Robustness test results for the alternative measure of the independent variable.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
<i>ESG_dis6</i>	<i>MISP_cross</i> 0.012*** (2.69)	<i>MISP_time</i> −0.029*** (−10.43)								
<i>ESGD</i>			0.224*** (4.31)	−0.364*** (−9.36)						
<i>EGrange4</i>					0.016*** (5.36)	−0.005** (−2.38)				
<i>ESGdis</i>							0.015*** (3.12)	−0.017*** (−5.38)		
<i>Disagreement</i>									0.148*** (3.98)	−0.385*** (−13.63)
<i>Age</i>	0.207** (2.24)	−0.054 (−0.88)	0.207** (2.24)	−0.058 (−0.96)	0.202** (2.20)	−0.066 (−1.08)	0.215** (2.33)	−0.071 (−1.15)	0.209** (2.27)	−0.057 (−0.95)
<i>ATO</i>	0.048** (2.33)	−0.022** (−1.98)	0.048** (2.32)	−0.022** (−2.00)	0.049** (2.39)	−0.023** (−2.10)	0.049** (2.35)	−0.023** (−2.10)	0.048** (2.31)	−0.020* (−1.90)
<i>ROA</i>	−1.173*** (−9.80)	0.600*** (9.38)	−1.170*** (−9.76)	0.595*** (9.31)	−1.176*** (−9.83)	0.602*** (9.36)	−1.178*** (−9.83)	0.606*** (9.42)	−1.175*** (−9.81)	0.604*** (9.59)
<i>Growth</i>	0.042*** (4.91)	−0.008* (−1.69)	0.042*** (4.94)	−0.008* (−1.69)	0.043*** (5.00)	−0.007 (−1.57)	0.042*** (4.88)	−0.007 (−1.50)	0.043*** (4.96)	−0.009* (−1.95)
<i>FAS</i>	0.185*** (3.26)	0.041 (1.28)	0.184*** (3.25)	0.041 (1.30)	0.180*** (3.18)	0.040 (1.23)	0.186*** (3.28)	0.038 (1.18)	0.184*** (3.23)	0.045 (1.46)
<i>Indep</i>	0.187** (2.24)	0.041 (0.80)	0.188** (2.25)	0.040 (0.77)	0.188** (2.26)	0.040 (0.78)	0.188** (2.26)	0.039 (0.76)	0.191** (2.29)	0.031 (0.61)
<i>Top3</i>	−0.608*** (−7.33)	−0.049 (−1.16)	−0.609*** (−7.36)	−0.042 (−1.00)	−0.603*** (−7.29)	−0.032 (−0.74)	−0.616*** (−7.44)	−0.029 (−0.67)	−0.607*** (−7.34)	−0.054 (−1.32)
<i>Turnover</i>	0.012*** (18.07)	0.006*** (17.86)	0.012*** (18.08)	0.006*** (18.00)	0.012*** (18.04)	0.006*** (18.12)	0.012*** (18.04)	0.006*** (18.08)	0.012*** (18.08)	0.006*** (18.04)
<i>Investor</i>	1.144*** (20.12)	0.169*** (6.40)	1.146*** (20.17)	0.166*** (6.29)	1.141*** (20.11)	0.171*** (6.42)	1.142*** (20.05)	0.173*** (6.49)	1.147*** (20.18)	0.161*** (6.19)
<i>Cons</i>	−0.988*** (−3.49)	0.217 (1.19)	−1.002*** (−3.55)	0.244 (1.33)	−0.986*** (−3.49)	0.225 (1.22)	−1.002*** (−3.54)	0.239 (1.29)	−0.995*** (−3.52)	0.236 (1.32)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	20626	20626	20626	20626	20626	20626	20626	20626	20626	20626
Adj R ²	0.6679	0.7589	0.6682	0.7593	0.6685	0.7567	0.6679	0.7572	0.6682	0.7636

Note: The t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

to contemporaneous companies in the same industry, while it is lower compared to long-term prices. The conclusions of this study are robust. Furthermore, this study employs multidimensional methods to compare ESG ratings, including Principal Component Analysis (PCA), range, interquartile range (IQR), and coefficient of variation (CV) methods to further measure ESG rating disagreement, re-estimates the model, and obtains results as shown in Table 4. The findings show that greater ESG rating disagreement leads to higher company stock valuation compared to contemporaneous companies in the same industry, and it is lower compared to long-term prices. The conclusions of this study are robust.

Specifically, this study first refers to work by Bissoondoyal-Bheenick et al. (2024), H. Chen and Wang (2025), adopting Principal Component Analysis (PCA) to assign weights to ESG scores derived from distinct institutions prior to computing ESG rating disagreement (*ESGD*).⁶ As detailed in Table 4, the coefficient of *ESGD* in Column (3) is 0.224, while that in Column (4) is −0.364. These findings confirm that our conclusions are robust. Second, this study uses extreme value differences, namely the difference between the highest and lowest ESG ratings provided by various institutions (*ESGrange4*) as a substitute for the explanatory variable (K. Hu, Li, and Li 2024). The estimates reported in Columns (5) and (6) demonstrate that even after replacing this independent variable, subsequent regression results remain robust. Furthermore, following the research of Tan and Pan (2023), ESG scores are first divided into quartiles of overall rankings by year, then the standard deviation of ratings from various institutions (*ESGdis*) is calculated to measure ESG rating disagreement. Empirical outcomes displayed in Columns (7) and (8) remain robust. Finally, following the approach of X. Liu, Dai, et al. (2024), the coefficient of variation (*Disagreement*) of ESG ratings received by company *i* in year *t* is used as a substitute indicator for ESG rating disagreement. The calculation formula is shown in Equation (10) below:

$$Disagreement_{it} = \frac{SD_{it}}{Mean_{it}} \quad (10)$$

Where SD_{it} denotes the standard deviation of rating scores that the company receives from various rating agencies, and $Mean_{it}$ represents the average of rating scores that the company receives from rating agencies. Upon re-estimation, empirical outcomes reported in Columns (9) and (10) maintain statistical significance.

Replacement of the Data Sample

Recently, China has actively engaged in formulating global ESG standards. Through absorbing and learning from international experience, it has established and improved domestic relevant policy and regulatory systems. The China Securities Regulatory Commission promulgated the “Guidelines for Investor Relations Management of Listed Companies” on April 11, 2022,⁷ incorporating ESG metrics into investor-firm dialogs while standardizing disclosure mandates. To mitigate confounding effects arising from this policy implementation, this study excludes the 2022 sample data and re-estimates the model, presenting outcomes in Table 5. Among these, Columns (1) and (2) show positive coefficients for *ESG_dis*, suggesting that as ESG rating disagreement increases, stock valuation becomes higher when compared horizontally with contemporaneous companies in the same industry. Conversely, Columns (3) and (4) exhibit significantly negative coefficients, implying that as ESG rating disagreement increases, stock valuation becomes lower in longitudinal comparison with long-term prices. The conclusions obtained remain robust.

Change in the Fixed Effects and Cluster Levels

Drawing on the work of Y. Yang et al. (2025), we verify the robustness of our results by varying the fixed effect and clustering settings. The development level of the industry or region may affect firm development. Therefore, we introduce the model with industry and year-by-city fixed effects, while retaining firm and year controls. Additionally, since different cluster level choices may impact

Table 5. Robustness test results of the alternative sample.

	(1)	(2)	(3)	(4)
	<i>MISP_{cross}</i>	<i>MISP_{cross}</i>	<i>MISP_{time}</i>	<i>MISP_{time}</i>
<i>ESG_dis</i>	0.010* (1.88)	0.014*** (2.61)	−0.023*** (−6.29)	−0.023*** (−6.28)
<i>Age</i>		0.199** (2.00)		0.027 (0.40)
<i>ATO</i>		0.054** (2.49)		−0.027** (−2.24)
<i>ROA</i>		−1.185*** (−8.87)		0.685*** (9.53)
<i>Growth</i>		0.033*** (3.81)		−0.010** (−2.13)
<i>FAS</i>		0.207*** (3.39)		0.066* (1.89)
<i>Indep</i>		0.174** (2.00)		0.046 (0.84)
<i>Top3</i>		−0.620*** (−6.85)		−0.110** (−2.33)
<i>Turnover</i>		0.012*** (17.44)		0.006*** (16.94)
<i>Investor</i>		1.145*** (18.30)		0.214*** (7.33)
<i>Cons</i>	−0.023*** (−5.75)	−0.963*** (−3.18)	0.200*** (74.61)	−0.007 (−0.04)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	17417	17417	17417	17417
Adj R ²	0.6306	0.6689	0.7625	0.7739

Note: The t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

statistical significance, this study uses alternative clustering levels, such as year-industry and year-city clustering. The results in Table 6 show that the conclusions obtained remain robust.

Lagged Control Variables

Since firm-level factors may affect ESG rating disagreement, this study follows the method of Y. Yang, Xie, and Qiang (2025) by employing lagged control variables from one period to reduce endogeneity. Table 7 reports the empirical results. Among these, the coefficients on *ESG_dis* are positive in Columns (1) and (2) and negative in Columns (3) and (4), indicating that as ESG rating disagreement increases, stock valuation becomes higher when compared horizontally with contemporaneous companies in the same industry, and stock valuation becomes lower in longitudinal comparison with long-term prices, pointing out that the basic conclusions of this study are valid.

Analysis Including Extreme Values

To study extreme values, this study re-estimates the full sample that only underwent sample screening procedures but did not remove extreme values, tabulating both descriptive statistics and main regression analyses in Tables 8 and 9. Among these, Table 8 indicates that descriptive statistics for most variables without removing extreme values are more volatile. Table 9 shows that although extreme values were not removed, the coefficients of *ESG_dis*, *Sen*×*ESG_dis*, and *KV*×*ESG_dis* remain significant and consistent with the direction of our main findings, indicating that they do not significantly impact the research conclusions.

Endogeneity Tests

First, factors such as company business models can precipitate disagreement among rating agencies regarding corporate ESG performance, thereby affecting stock mispricing. Second, the low pricing efficiency of China's capital market, coupled with the lack of standardized ESG rating criteria,

Table 6. Robustness test results for the change in the fixed effects and cluster levels.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	<i>MISP_{cross}</i>	<i>MISP_{time}</i>	<i>MISP_{cross}</i>	<i>MISP_{time}</i>	<i>MISP_{cross}</i>	<i>MISP_{time}</i>	<i>MISP_{cross}</i>	<i>MISP_{time}</i>
<i>ESG_dis</i>	0.024*** (5.10)	-0.028*** (-8.22)	0.023*** (4.40)	-0.026*** (-7.81)	0.024*** (5.21)	-0.028*** (-6.10)	0.024*** (5.69)	-0.028*** (-7.74)
<i>Age</i>	0.204** (2.22)	-0.059 (-0.96)	0.229** (2.28)	-0.019 (-0.28)	0.204*** (3.13)	-0.059 (-1.26)	0.204*** (3.52)	-0.059 (-1.52)
<i>ATO</i>	0.048** (2.33)	-0.022** (-2.06)	0.051** (2.24)	-0.017 (-1.49)	0.048*** (2.67)	-0.022* (-1.71)	0.048*** (2.96)	-0.022*** (-2.80)
<i>ROA</i>	-1.177*** (-9.82)	0.605*** (9.44)	-1.180*** (-9.13)	0.577*** (8.75)	-1.177*** (-10.56)	0.605*** (5.65)	-1.177*** (-11.96)	0.605*** (11.01)
<i>Growth</i>	0.043*** (4.97)	-0.008* (-1.72)	0.036*** (3.98)	-0.008 (-1.63)	0.043*** (4.74)	-0.008 (-1.25)	0.043*** (4.57)	-0.008 (-1.55)
<i>FAS</i>	0.182*** (3.20)	0.043 (1.35)	0.190*** (3.09)	0.040 (1.19)	0.182*** (3.85)	0.043 (1.52)	0.182*** (4.12)	0.043* (1.73)
<i>Indep</i>	0.190** (2.27)	0.037 (0.73)	0.179** (1.96)	0.037 (0.66)	0.190*** (2.85)	0.037 (0.89)	0.190*** (2.62)	0.037 (0.87)
<i>Top3</i>	-0.604*** (-7.29)	-0.043 (-1.02)	-0.605*** (-6.88)	-0.019 (-0.42)	-0.604*** (-9.62)	-0.043 (-0.99)	-0.604*** (-10.41)	-0.043 (-1.30)
<i>Turnover</i>	0.012*** (18.06)	0.006*** (17.99)	0.012*** (16.39)	0.006*** (16.96)	0.012*** (16.26)	0.006*** (11.31)	0.012*** (20.83)	0.006*** (18.04)
<i>Investor</i>	1.144*** (20.14)	0.170*** (6.38)	1.155*** (18.63)	0.165*** (5.72)	1.144*** (26.88)	0.170*** (6.51)	1.144*** (28.91)	0.170*** (8.39)
<i>Cons</i>	-0.991*** (-3.50)	0.226 (1.23)	-1.055*** (-3.43)	0.095 (0.48)	-0.991*** (-5.14)	0.226* (1.71)	-0.991*** (-5.47)	0.226* (1.93)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes				
Year×City FE			Yes	Yes				
Clustered at Year×Industry					Yes	Yes		
Clustered at Year×City							Yes	Yes
N	20626	20626	19572	19572	20626	20626	20626	20626
Adj R ²	0.6671	0.7576	0.6634	0.7608	0.6684	0.7585	0.6684	0.7585

Note: The t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

generates significant information noise and price distortions. Listed company management is more manipulative or has poorer disclosure quality, resulting in greater volatility in listed companies' ESG ratings (Ruan, Li, and Huang 2025). Third, ESG is non-mandatory information disclosure, and the disclosure of information is not a random process. Based on this, this study utilizes independent variables that are lagged one period, the instrumental variable method, a test for omitted variable bias, and the treatment effects model to alleviate endogeneity problems. Specifically, it is mainly divided into the following four parts.

(1) *Independent Variable Lagged One Period.* First, this study mitigates reverse causality problems by lagging explanatory variables. To alleviate potential bidirectional causal effects, we follow the approach of C. Zhang et al. (2024), and re-estimate the model with the one-period lagged ESG rating disagreement (*L.ESG_dis*). The empirical results are reported in Table 10. Across Columns (1) through (4), the coefficients on *L.ESG_dis* remain statistically significant. Moreover, the signs of the estimates align with the main regression, confirming that the conclusions remain robust.

(2) *Instrumental Variable Method.* This study uses the mean ESG rating disagreement among companies within the same city and year as instrumental variables. Companies in the same region often face similar external environments, and their ESG rating disagreement may be correlated. Meanwhile, the regional average level of ESG rating disagreement cannot directly affect corporate decision-making, thus satisfying the relevance and exogeneity conditions for instrumental variables.

Specifically, Table 11 reports results using two instrumental variables: the average ESG rating disagreement of firms within the same city and year (*ESGcity*), and the mean of other firms (*ESGcity1*). In the first-stage model estimation results, the coefficients of *ESGcity/ESGcity1* in Columns (1) and (4)

Table 7. Robustness test results of lagged control variables.

	(1)	(2)	(3)	(4)
	<i>MISP_{cross}</i>	<i>MISP_{cross}</i>	<i>MISP_{time}</i>	<i>MISP_{time}</i>
<i>ESG_dis</i>	0.021*** (4.22)	0.027*** (4.98)	−0.029*** (−8.61)	−0.020*** (−5.77)
<i>L.Age</i>		0.289** (2.57)		−0.011 (−0.16)
<i>L.ATO</i>		−0.000 (−0.01)		−0.032*** (−2.61)
<i>L.ROA</i>		0.359*** (2.87)		0.253*** (3.73)
<i>L.Growth</i>		−0.001 (−0.16)		−0.010** (−2.10)
<i>L.FAS</i>		0.155*** (2.71)		0.068** (1.97)
<i>L.Indep</i>		0.001 (0.01)		0.055 (0.97)
<i>L.Top3</i>		−0.103 (−1.10)		−0.025 (−0.50)
<i>L.Turnover</i>		0.003*** (4.11)		0.005*** (14.15)
<i>L.Investor</i>		0.345*** (5.95)		0.180*** (6.47)
Cons	−0.028*** (−6.88)	−1.066*** (−3.13)	0.183*** (67.38)	0.027 (0.13)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	20626	15319	20626	15319
Adj R ²	0.6297	0.6748	0.7480	0.6942

Note: The t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 8. Descriptive statistics of main variables without removing extreme values.

	N	Mean	SD	Min	Max
<i>MISP_{cross}</i>	21138	−0.007	0.456	−1.803	2.175
<i>MISP_{time}</i>	21138	0.155	0.310	−0.830	1.542
<i>ESG_dis</i>	21138	0.812	0.745	0.000	4.243
<i>ESG_dis6</i>	21138	0.940	0.717	0.000	4.243
<i>ESGD</i>	21138	0.115	0.086	0.000	0.434
<i>ESGrange4</i>	21138	1.377	1.284	0.000	7.000
<i>ESGdis</i>	21138	0.511	0.586	0.000	2.121
<i>Disagreement</i>	21138	0.076	0.085	0.000	0.530
<i>Sen</i>	20811	0.061	1.145	−3.022	6.956
<i>KV</i>	20491	0.409	0.229	0.001	2.033
<i>Age</i>	21138	2.929	0.308	1.386	4.159
<i>ATO</i>	21138	0.633	0.512	0.003	13.914
<i>ROA</i>	21138	0.053	0.046	−0.263	0.786
<i>Growth</i>	21138	0.246	2.967	−0.929	379.546
<i>FAS</i>	21138	0.206	0.153	0.000	0.954
<i>Indep</i>	21138	0.378	0.056	0.167	0.800
<i>Top3</i>	21138	0.486	0.154	0.006	0.983
<i>Turnover</i>	21138	6.452	5.566	0.053	80.288
<i>Investor</i>	21138	0.430	0.250	0.000	1.011

are statistically significant. Moreover, the Kleibergen-Paap rk LM test rejects the null hypothesis of instrumental variable under-identification at the 1% significance level, confirming that the instruments are identifiable. In the second-stage model, the coefficients of *ESG_dis* in Columns (2) and (5) are positive, while that in Columns (3) and (6) are negative, supporting Hypothesis H1. The Cragg-Donald Wald F Statistic exceeds 10, and the Kleibergen-Paap rk Wald F test statistic surpasses the critical value (16.38), ruling out weak instrument identification. Furthermore, this study follows the

Table 9. Baseline regression and mechanism analysis results for ESG disagreement and stock mispricing (without removing extreme values).

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	<i>MISP_{cross}</i>	<i>MISP_{time}</i>	<i>MISP_{cross}</i>	<i>MISP_{time}</i>	<i>MISP_{cross}</i>	<i>MISP_{time}</i>	<i>MISP_{cross}</i>	<i>MISP_{time}</i>
<i>ESG_dis</i>	0.022*** (4.23)	−0.029*** (−8.55)	0.025*** (5.07)	−0.028*** (−8.26)	0.011*** (2.87)	−0.027*** (−8.26)	0.010 (1.27)	−0.045*** (−8.71)
<i>Sen</i>					0.182*** (40.49)	0.016*** (5.47)		
<i>Sen</i> × <i>ESG_dis</i>					0.034*** (9.60)	−0.008*** (−3.44)		
<i>KV</i>							0.032* (1.79)	0.011 (0.97)
<i>KV</i> × <i>ESG_dis</i>							0.037** (2.38)	0.045*** (4.65)
<i>Age</i>			0.185** (2.04)	−0.042 (−0.70)	0.056 (0.82)	−0.070 (−1.15)	0.184** (1.99)	−0.050 (−0.82)
<i>ATO</i>			0.023* (1.66)	−0.014** (−2.03)	−0.002 (−0.23)	−0.017** (−2.49)	0.022 (1.56)	−0.012* (−1.80)
<i>ROA</i>			−0.951*** (−9.03)	0.483*** (8.70)	−1.455*** (−16.66)	0.477*** (8.65)	−1.037*** (−9.57)	0.470*** (8.42)
<i>Growth</i>			0.001 (0.54)	−0.001* (−1.71)	0.001 (1.56)	−0.001* (−1.71)	0.001 (0.53)	−0.001* (−1.71)
<i>FAS</i>			0.146*** (2.62)	0.054* (1.68)	0.015 (0.33)	0.041 (1.31)	0.150*** (2.66)	0.043 (1.35)
<i>Indep</i>			0.184** (2.26)	0.039 (0.78)	0.078 (1.21)	0.029 (0.59)	0.177** (2.15)	0.035 (0.69)
<i>Top3</i>			−0.627*** (−7.47)	−0.047 (−1.09)	−0.120* (−1.90)	−0.030 (−0.69)	−0.625*** (−7.23)	−0.046 (−1.04)
<i>Turnover</i>			0.011*** (17.31)	0.006*** (17.47)	0.012*** (25.04)	0.006*** (18.18)	0.011*** (17.05)	0.005*** (16.73)
<i>Investor</i>			1.187*** (20.34)	0.171*** (6.24)	0.721*** (15.81)	0.154*** (5.75)	1.172*** (19.76)	0.160*** (5.88)
<i>Cons</i>	−0.027*** (−6.66)	0.184*** (66.79)	−0.914*** (−3.27)	0.178 (0.98)	−0.484** (−2.31)	0.259 (1.42)	−0.915*** (−3.22)	0.200 (1.09)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	20626	20626	20626	20626	20294	20294	19962	19962
Adj R ²	0.6291	0.7434	0.6667	0.7532	0.7863	0.7537	0.6645	0.7515

Note: The t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively. The reduction in sample size here is due to the elimination of 512 singleton observations after the regression model.

research of Lee et al. (2022) and Cowx, Silva, and Yeung (2023) to estimate tF validity ratio within the IV framework. Specifically, it reports first-stage F-values, 5% and 1% adjustment factors, and the lower confidence bounds for the corresponding coefficients. The 95% and 99% CI lower bounds show consistency with the coefficient direction of the regression, reinforcing the robustness of the conclusions.

(3) *Test for Omitted Variable Bias.* Moreover, to eliminate endogeneity problems arising from omitted variables, this paper adopts the research of Oster (2019), Verner and Gyöngyösi (2020), Dantas, Merkley, and Silva (2023) to conduct coefficient stability analysis, adopting boundary testing methods for additional robustness testing of omitted variable bias. First, this study estimates regression without control variables to obtain the coefficient estimates of *ESG_dis* and R^2 , denoted as $\hat{\beta}$ and $\hat{R}^2$ respectively. Second, it considers a regression model including all control variables and again obtains coefficient estimates of *ESG_dis* and R^2 , recording them as $\tilde{\beta}$ and $\tilde{R}^2$ respectively. Finally, it uses safety factors to calculate R^2_{max} and estimates parameter bounds $\beta\Pi^*$ according to Equation (11). Table 12 presents these results. In the model involving the variable *MISP_{cross}*, using safety factors $\Pi = 1.3$ and $\Pi = 2$ to calculate the R^2_{max} estimated parameter bounds $\beta\Pi^*$ yields values of 0.040 and 0.050. Similarly, for the model involving the variable *MISP_{time}*, using safety factors $\Pi = 1.3$ and $\Pi = 2$ to calculate the R^2_{max} estimated

Table 10. Regression analysis results based on one-period lagged independent variable.

	(1)	(2)	(3)	(4)
	<i>MISP_{cross}</i>	<i>MISP_{cross}</i>	<i>MISP_{time}</i>	<i>MISP_{time}</i>
<i>L.ESG_dis</i>	0.023*** (3.95)	0.022*** (3.85)	−0.013*** (−3.24)	−0.013*** (−3.34)
<i>Age</i>		0.208* (1.78)		−0.077 (−0.98)
<i>ATO</i>		0.037 (1.59)		−0.007 (−0.60)
<i>ROA</i>		−1.203*** (−8.74)		0.615*** (8.74)
<i>Growth</i>		0.015 (1.51)		−0.011* (−1.95)
<i>FAS</i>		0.176*** (2.64)		0.032 (0.86)
<i>Indep</i>		0.192** (2.04)		0.046 (0.79)
<i>Top3</i>		−0.616*** (−6.06)		−0.077 (−1.55)
<i>Turnover</i>		0.018*** (18.98)		0.007*** (14.72)
<i>Investor</i>		1.276*** (17.82)		0.165*** (5.33)
<i>Cons</i>	−0.044*** (−10.45)	−1.082*** (−3.01)	0.113*** (40.73)	0.223 (0.94)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	15319	15319	15319	15319
Adj R ²	0.6708	0.7108	0.6845	0.6966

Note: The t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

parameter bounds $\beta\Pi^*$ yields values of -0.006 and -0.005 . The results indicate that the parameter bounds effectively preserve the sign of the main effect of *ESG_dis* previously recorded, indicating that even if omitted variable problems exist, they would not change the research inferences.

$$\beta^* = \tilde{\beta} - \frac{(\dot{\beta} - \tilde{\beta})(R_{max}^2 - \tilde{R}^2)}{(\tilde{R}^2 - \dot{R}^2)} \tag{11}$$

(4) *Treatment Effects Model.* This paper further utilizes the treatment effects model to discuss the issue of sample self-selection. Specifically, a dummy variable *Dum* is constructed based on the median value of ESG rating disagreement, and a two-step estimation method is used for empirical analysis.

In the first-stage model, two exclusion variables are incorporated: the average ESG disagreement of firms within the same city and year (*ESGcity*), and the mean of other firms (*ESGcity1*). Subsequently, the IMR is computed and included in the second-stage estimation, showing the results in Table 13. In the first stage, the coefficients for *ESGcity* and *ESGcity1* in Columns (1) and (4) are both positive. Moreover, the *IMR* coefficients in the second-stage estimation results are statistically significant in regressions on pricing efficiency at the cross-sectional dimension level and longitudinal time dimension level. This indicates that sample selection bias related to ESG rating disagreement does indeed exist, and it is necessary to discuss the issue of sample self-selection. Furthermore, the coefficients of *ESG_dis* remain significant at the 1% level, indicating that ESG rating disagreement tends to induce overvaluation in the cross-sectional dimension and undervaluation in the longitudinal time dimension, and its effect is not substantially affected by the sample self-selection problem.

Table 11. Endogeneity test based on 2SLS (using *ESGcity* and *ESGcity1* as instrument variables).

	First Stage	Second Stage	Second Stage	First Stage	Second Stage	Second Stage
	(1)	(2)	(3)	(4)	(5)	(6)
	<i>ESG_dis</i>	<i>MISP_{cross}</i>	<i>MISP_{time}</i>	<i>ESG_dis</i>	<i>MISP_{cross}</i>	<i>MISP_{time}</i>
<i>ESG_dis</i>		0.053*** (8.68)	−0.329*** (−76.75)		0.049*** (6.51)	−0.395*** (−69.22)
<i>ESGcity</i>	1.008*** (138.55)					
<i>ESGcity1</i>				0.838*** (90.85)		
<i>Age</i>	−0.072*** (−5.22)	−0.128*** (−12.30)	−0.050*** (−6.55)	−0.012 (−0.80)	−0.125*** (−11.64)	−0.031*** (−3.65)
<i>ATO</i>	−0.016* (−1.65)	−0.029*** (−3.67)	−0.008 (−1.46)	−0.023** (−2.09)	−0.030*** (−3.80)	−0.011* (−1.87)
<i>ROA</i>	0.404*** (4.10)	0.897*** (10.50)	0.921*** (16.71)	0.640*** (5.61)	0.957*** (10.79)	1.046*** (17.03)
<i>Growth</i>	−0.044*** (−3.58)	0.161*** (14.49)	0.039*** (5.29)	−0.078*** (−5.38)	0.156*** (13.63)	0.023*** (2.78)
<i>FAS</i>	−0.003 (−0.12)	−0.110*** (−5.54)	−0.102*** (−7.18)	−0.059** (−1.97)	−0.118*** (−5.76)	−0.105*** (−6.58)
<i>Indep</i>	0.046 (0.62)	0.382*** (6.74)	0.138*** (3.48)	0.089 (1.08)	0.381*** (6.55)	0.156*** (3.55)
<i>Top3</i>	−0.124*** (−4.10)	−0.007 (−0.30)	−0.301*** (−17.99)	−0.109*** (−3.23)	−0.009 (−0.38)	−0.307*** (−16.63)
<i>Turnover</i>	−0.005*** (−6.03)	0.009*** (14.33)	0.009*** (18.88)	−0.006*** (−6.85)	0.009*** (13.92)	0.008*** (15.72)
<i>Investor</i>	0.105*** (5.46)	0.316*** (21.01)	0.167*** (15.59)	0.103*** (4.77)	0.321*** (20.93)	0.168*** (14.22)
<i>Cons</i>	0.231*** (4.55)	−0.042 (−1.06)	0.505*** (17.86)	0.188*** (3.30)	−0.046 (−1.13)	0.504*** (16.20)
Kleibergen-Paap rk LM	6588.916 (<i>p</i> =.00)			4776.020 (<i>p</i> =.00)		
Cragg-Donald Wald F		14765.53	14765.53		7811.60	7811.60
Kleibergen-Paap Wald rk F		19195.92	19195.92		8253.84	8253.84
N	20626	20626	20626	19601	19601	19601
First-stage F-ratio	1.48e+04			7.82e+03		
5% Adjustment factor	0.006			0.008		
β 95% CI lower bound		0.040	−0.338		0.034	−0.406
1% Adjustment factor	0.006			0.008		
β 99% CI lower bound		0.037	−0.340		0.029	−0.410

Note: Parenthetical values are t-statistics for the first stage and z-statistics for the second stage. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 12. Omitted variable tests based on ESG rating disagreement and stock mispricing models.

	Simplified model		All model		R^2_{max}	Bounding values	R^2_{max}	Bounding values
	<i>MISP_{cross}</i>	<i>MISP_{cross}</i>	<i>MISP_{cross}</i>	<i>MISP_{cross}</i>	<i>MISP_{cross}</i>	<i>MISP_{cross}</i>	<i>MISP_{cross}</i>	<i>MISP_{cross}</i>
Outcome	$\hat{\beta}$	$\hat{R}^2$	$\tilde{\beta}$	$\tilde{R}^2$	$\Pi = 1.3$	$\beta^*_{\Pi=1.3}$	$\Pi = 2$	$\beta^*_{\Pi=2}$
Coefficient	0.021	0.6297	0.024	0.6684	0.8689	0.040	1	0.050
	Simplified model		All model		R^2_{max}	Bounding values	R^2_{max}	Bounding values
	<i>MISP_{time}</i>	<i>MISP_{time}</i>	<i>MISP_{time}</i>	<i>MISP_{time}</i>	<i>MISP_{time}</i>	<i>MISP_{time}</i>	<i>MISP_{time}</i>	<i>MISP_{time}</i>
Outcome	$\hat{\beta}$	$\hat{R}^2$	$\tilde{\beta}$	$\tilde{R}^2$	$\Pi = 1.3$	$\beta^*_{\Pi=1.3}$	$\Pi = 2$	$\beta^*_{\Pi=2}$
Coefficient	−0.029	0.7480	−0.028	0.7585	0.9861	−0.006	1	−0.005

Mechanism Tests

Columns (1) and (2) of Table 14 examine the moderating role of investor sentiment in the relationship between ESG rating disagreement and stock mispricing. The coefficient of *Sen*×*ESG_dis* in Column (1) is 0.033, while the coefficient in Column (2) is −0.008, indicating that investor sentiment positively moderates the impact of ESG rating disagreement on stock

Table 13. Endogeneity test based on the treatment effects model (using *ESGcity* and *ESGcity1* as exogenous variables).

	First Stage (1)	Second Stage (2)	Second Stage (3)	First Stage (4)	Second Stage (5)	Second Stage (6)
	<i>Dum</i>	<i>MISP_{cross}</i>	<i>MISP_{time}</i>	<i>Dum</i>	<i>MISP_{cross}</i>	<i>MISP_{time}</i>
<i>ESG_dis</i>		0.051*** (5.62)	−0.027*** (−7.43)		0.032*** (3.03)	−0.028*** (−7.28)
<i>ESGcity</i>	2.179*** (30.14)					
<i>ESGcity1</i>				0.225*** (3.57)		
<i>IMR</i>		0.049*** (6.14)	0.009*** (3.21)		0.064*** (7.29)	0.011*** (3.42)
<i>Age</i>	−0.111** (−1.99)	−0.137*** (−6.62)	0.003 (0.73)	−0.101* (−1.79)	−0.137*** (−6.48)	0.004 (0.76)
<i>ATO</i>	0.047 (1.04)	−0.069*** (−3.75)	0.004 (1.00)	0.022 (0.49)	−0.069*** (−3.71)	0.005 (1.18)
<i>ROA</i>	1.269*** (3.39)	0.921*** (6.17)	0.746*** (19.40)	0.988*** (2.63)	0.986*** (6.43)	0.763*** (19.68)
<i>Growth</i>	−0.069* (−1.79)	0.146*** (11.67)	0.014*** (3.30)	−0.088** (−2.31)	0.141*** (11.01)	0.013*** (3.04)
<i>FAS</i>	−0.197 (−1.50)	−0.127*** (−2.76)	−0.001 (−0.07)	−0.267** (−2.07)	−0.143*** (−3.05)	−0.001 (−0.11)
<i>Indep</i>	−0.080 (−0.29)	0.362*** (3.57)	0.036 (1.48)	0.204 (0.74)	0.372*** (3.60)	0.037 (1.54)
<i>Top3</i>	−0.310*** (−2.60)	0.020 (0.41)	−0.093*** (−8.74)	−0.334*** (−2.82)	0.013 (0.27)	−0.094*** (−8.63)
<i>Turnover</i>	−0.019*** (−7.56)	0.009*** (10.80)	0.002*** (9.25)	−0.020*** (−7.82)	0.009*** (10.48)	0.002*** (8.58)
<i>Investor</i>	0.434*** (5.89)	0.343*** (11.90)	0.073*** (10.63)	0.497*** (6.85)	0.346*** (11.88)	0.072*** (10.37)
<i>Cons</i>	−1.283*** (−4.45)	−0.007 (−0.08)	0.113*** (6.26)	−1.036*** (−3.34)	0.011 (0.14)	0.114*** (6.23)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
N	20621	20621	20621	19596	19596	19596
R ² /Adj R ²	0.3296	0.1173	0.7333	0.2687	0.1186	0.7348

Note: Parenthetical values are z-statistics for the first stage and t-statistics for the second stage. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

mispricing. This may be because individual investors are more prone to pronounced emotional biases in investment decisions and fail to achieve complete rationality, as they are influenced by market sentiment to change their investment decisions. Specifically, under high market sentiment, investors reduce their scrutiny of real enterprise information, providing managers with greater incentives to conceal the suboptimal investment performance of their actual industries (H. Wang and Zhang 2021). Moreover, when market investor sentiment is relatively optimistic, it leads to increased market demand, which in turn results in high turnover rates. During this process, compared to contemporaneous companies in the same industry, stock prices become increasingly overvalued, while compared to long-term valuation, stock prices are also more significantly undervalued. Individual investors are prone to generating relatively extreme optimistic or pessimistic emotions, thereby causing stock pricing to be too high or too low; however, such excessively high or low pricing is difficult to sustain (C. He et al. 2021). When investor sentiment is low, pessimists constitute a larger proportion of the corporate shareholding group. These pessimistic investors scrutinize information sources more strictly and hold pessimistic beliefs. Meanwhile, they evaluate corporate investment behavior in a more cautious, systematic, and detailed manner, being less affected by market noise (H. Wang and Zhang 2021). Confronted with high ESG disagreement, most investors adopt more cautious strategies (H. Wang et al. 2024), thereby attenuating the impact of ESG rating disagreement on stock mispricing. Therefore, investor sentiment exerts a positive moderating effect.

Table 14. Mechanism test results of investor sentiment and information disclosure quality.

	(1)	(2)	(3)	(4)
	<i>MISP_{cross}</i>	<i>MISP_{time}</i>	<i>MISP_{cross}</i>	<i>MISP_{time}</i>
<i>ESG_dis</i>	0.010*** (2.82)	−0.027*** (−8.23)	0.010 (1.30)	−0.043*** (−8.53)
<i>Sen</i>	0.187*** (42.40)	0.016*** (5.44)		
<i>ESG_dis</i> × <i>Sen</i>	0.033*** (10.27)	−0.008*** (−3.34)		
<i>KV</i>			0.031* (1.79)	0.008 (0.72)
<i>ESG_dis</i> × <i>KV</i>			0.035** (2.26)	0.041*** (4.29)
<i>Age</i>	0.062 (0.90)	−0.085 (−1.39)	0.203** (2.16)	−0.060 (−0.98)
<i>ATO</i>	0.020 (1.22)	−0.024** (−2.25)	0.050** (2.33)	−0.019* (−1.72)
<i>ROA</i>	−1.887*** (−20.65)	0.591*** (9.31)	−1.277*** (−10.66)	0.584*** (9.15)
<i>Growth</i>	0.071*** (9.03)	−0.004 (−0.93)	0.042*** (4.79)	−0.010** (−2.17)
<i>FAS</i>	0.064 (1.49)	0.034 (1.08)	0.181*** (3.16)	0.030 (0.94)
<i>Indep</i>	0.078 (1.19)	0.026 (0.51)	0.182** (2.16)	0.033 (0.63)
<i>Top3</i>	−0.106* (−1.71)	−0.028 (−0.65)	−0.601*** (−7.07)	−0.041 (−0.96)
<i>Turnover</i>	0.014*** (26.96)	0.006*** (18.86)	0.012*** (17.87)	0.006*** (17.33)
<i>Investor</i>	0.677*** (15.37)	0.153*** (5.87)	1.131*** (19.64)	0.161*** (6.06)
<i>Cons</i>	−0.511** (−2.42)	0.301 (1.64)	−0.993*** (−3.45)	0.228 (1.24)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	20294	20294	19962	19962
Adj R ²	0.7890	0.7589	0.6666	0.7567

Note: The t-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Columns (3) and (4) in Table 14 further reveal the role of information disclosure quality in the relationship between ESG rating disagreement and stock mispricing. Specifically, the coefficient of *KV* × *ESG_dis* in Column (3) is 0.035, while that in Column (4) is 0.041. The results show that information disclosure quality exerts a negative moderating effect on the relationship between ESG rating disagreement and cross-sectional dimension mispricing, while exerting a positive moderating effect on longitudinal mispricing. This may be because when information disclosure quality is inadequate, the presence of more inconsistent information leads to greater divergence in interpretations among different rating agencies. Conversely, high-quality disclosure helps reduce information asymmetry (Feng, Guo, and Zeng 2024). Investors can promptly and comprehensively access relevant information about listed companies and incorporate it into their investment decisions, thereby reducing stock valuation biases caused by information opacity. Therefore, in this case, information disclosure quality can alleviate the overestimation of cross-sectional pricing caused by ESG rating disagreement. In the long term, as the quality of disclosed information improves, the potential problems and risks of enterprises gradually become clear. In this situation, investors may pay more attention to negative information that could affect company performance, while neglecting the company's long-term progress in environmental, social responsibility, or governance aspects, thereby suppressing valuations in longitudinal mispricing. In addition, investors' reactions to corporate ESG information are related to information quality. As information disclosure quality improves, intensified rating disagreement may lead to market pricing lags and errors, especially in longitudinal time series. This dynamic

correction process means that the market may overreact to certain negative information and underestimate a firm's long-term sustainable development potential, thereby reducing stock price performance in the longitudinal dimension. Therefore, in this situation, as corporate information disclosure quality improves, although it can alleviate the positive impact of ESG rating disagreement on cross-sectional stock pricing, due to investors' lagged cognitive correction of long-term value and increased corporate risk exposure, the negative impact of ESG rating disagreement on stock prices in the longitudinal dimension is strengthened. In sum, as information disclosure quality improves, the impact of ESG rating disagreement on cross-sectional dimension pricing overvaluation decreases, while its impact on longitudinal dimension pricing undervaluation strengthens. Particularly, the coefficient of *ESG_dis* in Column (3) failed to reach statistical significance, which may be because the interaction term contains information from the independent variable. After introducing the interaction term, there will inevitably be changes in the main effect coefficient or sign, which is a common empirical issue. Additionally, the direction and intensity of the moderating effect may also cause the main effect to be offset or masked, ultimately making the main effect no longer significant. This phenomenon is also widely reflected in existing research (Heese, Pérez-Cavazos, and Peter 2022; L. Xue et al. 2025; P. Zhang, Ma, and Chi 2025; Zhong et al. 2025; Zhuang et al. 2025).

Further Analysis

Furthermore, this study examines the relationship between ESG rating adjustment directions and stock mispricing, constructing the directional changes in ESG ratings. Existing research on ESG rating changes examines adjustments to individual ESG rating indicators, exploring their impact on market returns (Bachner 2025; Cauthorn et al. 2023; Galema and Gerritsen 2025; Shanaev and Ghimire 2022), earnings management (Xiao, Ge, and Yu 2024), and corporate financial performance (Chou, Yan, and Huang 2025). H. Wang et al. (2024) introduced the concept of ESG improvement potential, which depends on whether a company's ESG rating for the current year equals or exceeds its rating from the previous year. Additionally, J. Zhang et al. (2024) studied the impact of ESG rating changes on fund holdings, using the average changes of multiple ESG ratings but did not consider the adjustment direction of individual ESG rating movements. In summary, prior studies have largely concentrated on the impact of individual ratings and how they affect various indicators of market outcomes across different time ranges. However, there has been no research on the impact of ESG rating adjustment directions on stock mispricing, nor has a methodology been established to effectively quantify ESG rating adjustment directions based on multiple ESG ratings. Therefore, this study starts from individual ESG ratings, separately setting directional indicators for individual ESG rating adjustments and studying their impact on stock market mispricing.

Based on this, to construct individual ESG rating adjustment indicators, this study adopts the methodology employed by scholars (Cauthorn et al. 2023; Galema and Gerritsen 2025; H. Wang et al. 2024), selecting the changes in ESG ratings and their dummy variables for setting. Specifically, for the individual rating Bloomberg, $\Delta Rating_Bloomberg$ represents the year-on-year ESG score differential, while *Moti_Bloomberg* is a binary variable assigned a value of 1 for rating upgrades and 0 for downgrades. First, subsamples are established based on groupings of changes in each ESG rating where ESG ratings have upgrades or downgrades, and the model is re-estimated. Second, dummy variables for changes in ESG ratings replace the independent variables and the model is re-estimated. Specifically, the empirical results for the impact of ESG rating changes from six individual institutions—Huazheng, Wind, SynTao Green Finance, SusallWave, Bloomberg, and FTSE Russell—on stock mispricing are presented in Tables B1 through B6 of Appendix B.

Among these, for FTSE Russell and SusallWave ratings, their own changes do not impact stock mispricing. This may be attributed to several factors: on the one hand, the data samples for most ESG ratings are not very large; on the other hand, most ESG ratings have not undergone adjustments, which can be inferred from the sample sizes in the results. In samples where Wind ESG ratings are upgraded, ESG ratings have a positive influence on stock prices in the cross-sectional dimension. In

samples where Bloomberg ESG ratings are upgraded, ESG ratings have a negative effect on stock prices in the longitudinal time dimension. Particularly, for Huazheng, whether in samples where ESG ratings are upgraded or downgraded, Huazheng ESG ratings lead to stock price overvaluation in the cross-sectional dimension and undervaluation in the longitudinal time dimension. That is, as Huazheng ESG rating improves, the company's stock valuation consistently becomes higher compared to contemporaneous companies in the same industry, and lower compared to long-term prices. This may be explained by the fact that when an enterprise's own ESG rating improves, due to market overreaction to ESG ratings, there is immediate short-term recognition of the company's improvements in social responsibility, environmental protection, and governance structure, leading to higher recognition in investors' minds and thereby increasing its stock price valuation. However, due to challenges in the company's actual operations, long-term performance has not been able to fully support these high valuations. These expectations may not be fully realized, and since the market remains unclear about the long-term impacts of ESG improvements, stock prices may reflect lag effects or conservative future expectations, subsequently resulting in a negative relationship.

Market reactions to Wind and Bloomberg after ESG rating upgrades may be influenced by their respective rating methodologies, market coverage, and different interpretations of ESG factors. Specifically, the Bloomberg ESG information disclosure score is derived from standardized metrics sourced from annual filings, sustainable disclosures, company websites, and other supplementary channels (Agapova, King, and Ranta 2025). Wind constructs its indicators through publicly disclosed information, news sentiment, and regulatory penalty databases, comprising two components: "management practices" and "controversial events."⁸ Additionally, the coefficient of *Moti_Huazheng* also shows a similar direction, indicating that compared to rating downgrades, its own ESG rating upgrades are more likely to cause stock price overvaluation in the cross-sectional dimension and undervaluation in the longitudinal time dimension. This result suggests that compared to ESG rating downgrades, Huazheng ESG rating upgrades exhibit certain bias in their impact on stock prices, further supporting the view that optimistic sentiment triggered by rating upgrade bias often exceeds cautious sentiment triggered by rating downgrade bias. However, for samples with SynTao Green Finance ESG rating upgrades, ESG ratings have a positive impact on stock prices in the longitudinal time dimension. This may be because SynTao Green Finance's ESG rating framework includes 14 core issues and customizes indicator weights for 51 industries respectively, setting weights according to industry characteristics to better capture sustainable development performance across different industries. Its ESG rating upgrades may stimulate positive expectations from the market and investors regarding enterprises' future performance, directly driving positive changes in stock prices in the longitudinal time dimension.

Discussion

Based on Chinese listed companies spanning 2015 through 2022, this paper examines the effect of ESG rating disagreement on stock mispricing. The benchmark results indicate that as ESG rating disagreement increases, the company's stock valuation is higher compared to others in the same industry, while it is lower compared to long-term prices. This is primarily because when investors are confronted with ESG rating disagreement, they tend to select the highest ESG rating as the company's true ESG performance. This leads to overestimation of the firm's ESG performance, thereby causing stock overvaluation. However, in the long term, this overvaluation becomes unsustainable, ultimately reducing long-term stock valuation. The findings of this study are consistent with existing literature proposing that ESG rating disagreement exerts a noise effect. These studies emphasize that ESG rating disagreement can cause information overload and cannot provide effective ESG information for investor decision-making. Specifically, it exacerbates the difficulty of interpreting the ESG information of enterprises (Anselmi and Petrella 2025; H. Wang et al. 2024; K. Hu, Li, and Li 2024; X. Lin, Zhu, and Meng 2023; Ruan, Li, and Huang 2025; X. Liu, Yang, et al. 2024; X. Zhang, Sun, and Gao 2025). Building on existing research, this study explores the impact of ESG rating disagreement on stock

mispricing, better demonstrating the noise effect of ESG rating uncertainty and providing scientific recommendations for regulatory frameworks.

In addition to baseline regression results, this study explores the mechanisms through which ESG rating disagreement influences stock mispricing. Mechanism tests reveal that investor sentiment and information disclosure quality significantly moderate this relationship. First, investor sentiment has a positive moderating influence on the relationship between ESG rating disagreement and stock mispricing. Under high market sentiment, investors reduce their scrutiny of real enterprise information, giving managers a greater incentive to conceal the poor investment returns of their actual industries. Moreover, when market investor sentiment is relatively optimistic, it leads to increased market demand, which in turn results in high turnover rates. During this process, compared to contemporaneous companies in the same industry, stock prices become increasingly overvalued, while compared to long-term valuation, stock prices are also more significantly undervalued. When investor sentiment is low, pessimists constitute a larger proportion of the corporate shareholding group, and this pessimistic group scrutinizes information sources more strictly and is less affected by market noise. When confronted with pronounced divergence in ESG ratings, most investors tend to embrace conservative approaches, thereby attenuating the effect of ESG rating disagreement on stock mispricing. Second, information disclosure quality has a negative moderating impact on the relationship between ESG rating disagreement and cross-sectional mispricing, while having a positive moderating impact on the relationship with longitudinal time dimension mispricing. The possible reason is that when information disclosure quality is high, investors can timely and comprehensively grasp relevant information about listed companies and incorporate it into investment decisions, reducing stock valuation deviations caused by information opacity. However, over the long term, as disclosure information quality improves, the company's potential problems and risks also gradually become clear. In this situation, investors may pay more attention to negative information that could affect company performance while ignoring the company's long-term progress in environmental, social responsibility, or governance aspects. Intensified rating disagreement may lead to market pricing lags and errors, where the market may overreact to negative information and underestimate the company's long-term sustainable development potential. Therefore, in this situation, information disclosure quality can weaken the overvaluation effect of ESG rating disagreement on cross-sectional pricing while strengthening the undervaluation in the longitudinal time dimension. Further analysis indicates that changes in individual ESG ratings also affect stock mispricing. Specifically, in samples where Wind ESG ratings are upgraded, ESG ratings show a positive relationship with stock prices in the cross-sectional dimension. In samples where Bloomberg ESG ratings are upgraded, ESG ratings show a negative relationship with stock prices in the longitudinal time dimension. Particularly, for Huazheng ESG ratings, whether in subsamples where ESG ratings are upgraded or downgraded, improvements in Huazheng ESG ratings lead to stock price overvaluation in the cross-sectional dimension and undervaluation in the longitudinal time dimension. The coefficient on the $\Delta Rating_Huazheng$ also shows a similar direction, indicating that compared to rating downgrades, its own ESG rating upgrades more significantly affect stock price overvaluation in the cross-sectional dimension and undervaluation in the longitudinal time dimension. This finding suggests that, compared to ESG rating downgrades, Huazheng ESG ratings upgrades show a certain bias in their impact on stock prices. In addition, for samples with ESG rating upgrades from SynTao Green Finance, ESG ratings demonstrate a positive relationship with stock prices in the longitudinal time dimension.

This paper makes several theoretical contributions to the literature on the economic consequences of ESG rating disagreement: First, this research finds that ESG rating disagreement can affect stock mispricing. Specifically, ESG rating disagreement leads to overvaluation of stock prices in the cross-sectional dimension and undervaluation in the longitudinal time dimension. This enriches existing research on the relationship between the two and provides empirical support for the “noise effect.” Second, this study introduces moderating variables to refine the influence mechanism. By exploring the moderating effects of investor sentiment and information

disclosure quality, this study further clarifies how ESG rating disagreement influences stock mispricing under different circumstances. Furthermore, the adjustment of individual ESG ratings also influences stock mispricing. This analysis offers a more comprehensive framework for understanding how ESG rating disagreement affects stock mispricing, which is conducive to a deeper understanding of the economic effects of ESG rating disagreement. Additionally, the findings of this study hold significant implications for investors and regulatory authorities: for investors, ESG rating disagreement often reflects an overly optimistic bias. By comparing ESG ratings from multiple agencies and analyst reports, investors need to integrate ESG information with traditional financial metrics to mitigate the negative impact of ESG rating disagreement. For regulatory authorities, the research findings emphasize standardization to improve the quality and comparability of ESG reporting. Strengthening accountability through both incentives and disciplinary measures can effectively reduce biased environmental, social, and governance reporting.

There are still some limitations in this study, which should be taken into account in future research. First, our sample is limited to Chinese listed firms, which may to some extent restrict the application scope of the research results in other institutional contexts. Future research may explore how ESG rating disagreement influences stock mispricing for foreign firms, in order to compare the influence of ESG rating disagreement across domestic and international contexts. Second, our analysis relies solely on disagreement in overall ESG scores. Future research will utilize disagreement in individual E, S, and G indicators to explore the impact of ESG rating disagreement on stock mispricing. Finally, this paper focuses on stock market performance, but companies also rely on consumers and suppliers. Consequently, future studies may explore how ESG rating disagreement affects the behavior of these groups.

Conclusions and Policy Implications

ESG rating disagreement has gradually become a significant factor in investment decisions. This paper extends the research on ESG rating disagreement to the aspect of stock mispricing. Specifically, utilizing a dataset of Chinese listed companies from 2015 to 2022, the study effectively reveals the significant impact of ESG rating disagreement on stock mispricing. Firstly, ESG rating disagreement affects stock mispricing. That is, as ESG rating disagreement increases, it causes overvaluation in the cross-sectional dimension and undervaluation in the longitudinal time dimension. Secondly, the mechanism test results reveal that investor sentiment and information disclosure quality significantly moderate the relationship between ESG rating disagreement and stock mispricing. Finally, adjustments to individual ESG ratings also affect stock mispricing. The research findings provide supportive evidence for studies on ESG rating disagreement's impact on corresponding asset pricing. They offer guidance for policymakers on optimizing ESG management and for corporate managers on enhancing sustainable development performance, and provide valuable insights for other emerging markets:

First, standardize ESG disclosure and evaluation systems, while introducing corresponding reward and punishment mechanisms. This study demonstrates that both ESG rating disagreement and adjustments to individual ESG ratings can impact stock mispricing. The government should unify the disclosure standards for sustainable development reports and standardize the scope and weighting of ratings. To ensure the reliability of data underlying ESG ratings, the government should establish a public platform for ESG information disclosure. Additionally, big data and artificial intelligence technologies should be utilized to enhance monitoring and analytical capabilities for ESG information, facilitating investors' access to and comparison of ESG information of different enterprises.

Second, strengthen investor education and improve their understanding of ESG. This research demonstrates that investor sentiment plays a key role in how ESG rating disagreement affects stock mispricing. Therefore, it is necessary to enhance ESG investors' accurate understanding of ratings and reduce their reliance on rating agencies. Financial institutions can promote investor education by organizing investor education seminars and conducting ESG investment training, thereby helping investors gradually build their understanding of ESG.

Third, establish accountability mechanisms for information disclosure and strengthen the oversight of corporate reporting practices. This study demonstrates that information disclosure quality significantly moderates the relationship between ESG rating disagreement and stock mispricing. Regulatory authorities should actively promote the ESG information disclosure practices of companies, requiring enterprises to regularly and comprehensively disclose their ESG performance. Authorities should also review these disclosures regularly. Additionally, a strict accountability mechanism should be introduced. Enterprises that fail to disclose ESG information as required or deliberately conceal it should be subject to corresponding penalties. At the same time, strengthening oversight of information disclosure and implementing incentive-based mechanisms will encourage companies to take ESG reporting more seriously, thereby enhancing the sustainable development of the market.

Highlights

1. ESG rating disagreement leads to stock mispricing, as evidenced by the overvaluation in the cross-sectional dimension and undervaluation in the longitudinal time dimension.
2. Investor sentiment positively moderates the relationship between ESG rating disagreement and stock mispricing.
3. Information disclosure quality negatively moderates the relationship between ESG rating disagreement and cross-sectional mispricing, while positively moderating the relationship between ESG rating disagreement and longitudinal mispricing.
4. Changes in individual ESG ratings also affect stock mispricing.

Notes

1. With the exception of the Japanese market data, which has a cutoff date of March 31, 2022, the remaining data in this report have a cutoff date of December 31, 2021.
2. Among these, this phenomenon is not caused by selection bias. A more common situation is that investors observe that some of a company's ESG ratings are low and subsequently inquire about the company. However, listed companies often respond to investors by highlighting other higher ratings of their own and disclosing awards received for ESG governance, aiming to gain favor from investors under the condition of ESG information asymmetry.
3. In this study, the cross-section refers to the company level, thus "company-level mispricing" and "cross-sectional dimension mispricing" are consistent in meaning. "Industry-level mispricing" and "longitudinal time dimension mispricing" capture the same phenomenon. Specifically, the definition of $v(\theta_{it}; a_{it}) - v(\theta_{it}; a_j)$ represents the difference between valuations derived from contemporaneous industry peers and those based on long-term industry peers, and is referred to as "time-series industry mispricing," indicating that it has attributes in both the time and industry dimensions.
4. This study uses Stata for empirical analysis, including descriptive statistics, dependent variable tests, and benchmark regression results.
5. This data period is selected based on Chinese listed companies' annual report disclosure timing. Chinese listed firms are required to disclose annual reports within four months from the end of each accounting year, specifically from January 1 to April 30 of each year.
6. Before conducting PCA, KMO and Bartlett tests are utilized to verify data suitability. The results show that the KMO statistics for all four distinct ESG ratings are greater than 0.5, as is the KMO value for the composite score. The KMO value for Huazheng is 0.8037, for Wind is 0.7781, for SynTao Green Finance is 0.7590, and for SusallWave is 0.7438. Regarding the composite score, the KMO value is 0.7670, the Bartlett p-value is 0.00, and the chi-square value is 4752.804.
7. https://www.gov.cn/xinwen/2022-02/09/content_5672688.htm.
8. Among these, management practice indicators include carbon emissions, employee benefits, and board structure. The controversial events are dynamically assessed based on news sentiment, legal proceedings, and regulatory penalties.

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